

Is Generative AI Displacing Arabic? Perceived Change in Arabic Use Among Lebanese Arabic–English Bilinguals

Hadi Ghazi
Lebanese University
hadigghazi@gmail.com

Abstract: Generative AI tools work better in English than in Arabic, so Arabic speakers who also know English may turn to English when they use them. In an anonymous online survey in Arabic, 105 Lebanese Arabic–English bilinguals who use AI regularly rated, for eight areas of daily life, whether they now use Arabic less or more because of AI. About two thirds reported using less Arabic in at least one area, most often in work or study (half of respondents), in writing messages and in following Arabic content, and least often with family and friends and in religion. Work or study came first, as theories of language shift predict, but formal Arabic (Fusha) did not decline significantly more than the spoken dialect. The more often respondents wrote to AI in English, the larger the decrease they reported; how much they used AI added nothing once this was taken into account. Simple machine-learning models identified, with moderate accuracy, who reports a decrease in formal areas, in work or study and in self-talk and own notes, but AI-use information added no significant accuracy to background characteristics such as age and education. Respondents who felt that AI reduced their use of Arabic felt the same about social media, consistent with AI continuing a shift that social media began. The findings describe perceived change; tracking actual language use over time is the next step.

Index Terms: Arabic, diglossia, generative AI, language shift, machine learning, perceived language change, survey

I. INTRODUCTION

Generative AI tools perform less well in most languages than in English, Arabic included [1]. Tool-calling accuracy drops when the same tasks are prompted in Arabic [2], and four chatbots answered the same infectious-disease questions less well in Arabic than in English [3]. Bilinguals who use these tools daily may therefore do more of their AI-assisted tasks in English. Arabic was losing formal domains to English, and in Lebanon also to French, before generative AI. English dominates higher education in the UAE [4], where students at English-medium universities report weaker academic Arabic [5]; Qatar University students rate English more useful than Arabic for scientific and professional communication [6]; and in Lebanon, where English has spread alongside French [7], a small interview study found English used across education and professional life [8].

Research on generative AI and its users’ language skills has looked almost entirely at second-language learning, e.g., [9]. The closest work on Arabic records which language Tunisian users choose with ChatGPT [10] but does not ask whether that choice affects their Arabic elsewhere. To our knowledge,

no study has asked Arabic speakers whether they attribute changes in their own Arabic use to AI.

This study asks Lebanese Arabic–English bilinguals in which domains, and in which direction, they report that AI has changed their use of Arabic (RQ1); whether AI-use intensity, the English share of AI use and the perceived Arabic–English quality gap predict that change (RQ2); whether the change goes with perceived difficulty using Arabic without AI and mediates the link from AI use (RQ3); and whether AI-use information improves out-of-sample prediction of who reports a decrease beyond background characteristics (RQ4). The outcome is the change that respondents attribute to AI, not a measure of language loss.

II. BACKGROUND AND HYPOTHESES

Domains and diglossia. Language use is organised by domains such as family, friendship, religion, education and work, and in a community undergoing language shift a dominant language displaces a weaker one domain by domain, starting with public and formal domains [11], [12]. Such shift unfolds over generations; this study examines its individual counterpart, a change in the language a person uses in each domain. Arabic is diglossic: Fusha (Modern Standard Arabic) carries the formal, written functions and the spoken dialect, here mostly Lebanese Arabic, the everyday ones [13]. If AI moves formal tasks into English, Fusha should be affected first. Hence **H1**: net AI-attributed change is negative for Fusha (H1a) and for work or study (H1b); and **H2**: the decrease follows the order work/study > personal matters > family and friends (H2a) and is larger for Fusha than for the everyday domains of self-talk and own notes, personal matters, and family and friends (H2b).

Why AI could matter. Adult first-language attrition is usually explained by reduced use of the first language and interference from an active second language [14], [15]. It has been studied mainly in migrants; less is known about speakers at home, where the first language stays in daily use. Students at English-medium universities in Turkey showed no deficit on Turkish tests, although their self-rated Turkish writing fell more often than that of Turkish-medium peers [16]. Generative AI could add a new route to reduced use, following an earlier one: social networking sites moved young Arabs’ informal writing into Latin script (Arabizi), often mixed with English, as

documented in Saudi Arabia and Jordan [17], [18]. Once access to a technology is equal, inequality moves from access to what people can do with it [19]; with AI, what bilinguals can do also depends on the language they use. Hence **H3**: higher AI-use intensity (H3a), a larger share of AI use in English (H3b) and a larger perceived quality gap (H3c) predict a larger AI-attributed decrease; **H4**: a larger decrease goes with more perceived difficulty using Arabic without AI; and **H5**: the decrease mediates the link from AI-use intensity to that difficulty.

Measuring attributed change. Wirtz and Pickl [20] showed that people can rate in an online survey how a major life event changed their language use, on a scale with a true zero (“no change”). Such retrospective attributions are only partly accurate: perceived change after life events correlated only about $r = 0.2$ with measured personality change, and also with change that began before the event [21]. This study adapts the format to AI and reads the results as attributions.

III. METHOD

A. Design, Procedure and Participants

An anonymous seven-page Google Form in Arabic (its one optional open question is not analysed here) received responses from 30 September to 4 October 2026, through university WhatsApp groups, friends and their contacts, Instagram stories and LinkedIn posts. Before launch, a speech-language pathologist reviewed the form and approved it without changes; the Arabic wording is the instrument (Appendix). The first page described the survey as voluntary, anonymous and reported as group results, and respondents confirmed being 18 or older and agreeing to take part. A screening question required Arabic as first language, use of English for study or work, and weekly use of AI tools (the form named ChatGPT, Gemini and Claude) for six months or more.

All 105 respondents passed the screening question (two began regular AI use in 2026 and may not meet the six-month criterion), and no submission duplicated another. Of these, 101 grew up in Lebanon, 96 live there and 12 had moved country since 2022; 82 were under 25 and 23 were 25–44; 60 were women; 55 held or were studying for a master’s degree or higher; 45 studied or worked in computing and 22 in engineering; 69 had studied school science in English and 35 in French; 79 used AI several times a day and 37 always wrote to it in English. All 105 form the primary sample; a core sample that holds migration constant (grew up and live in the Arab region, no move since 2022; $n = 89$) is a sensitivity check.

B. Measures

The core grid asked: “Because of AI, do you now do each of these less or more?” (Arabic in the Appendix), comparing now with before regular AI use, from *much less* (−2) through *no change* (0) to *much more* (+2). Its eight rows (Fig. 1; Appendix) mix domains in Fishman’s sense (work or study, family and friends, religion), activities (messages, content, self-talk and notes) and one variety (Fusha); all are called

domains for brevity. A preceding grid, which defined Arabic as one’s dialect or Fusha, recorded the language each activity is mostly done in today; a domain marked not applicable there was set to missing (removing seven decreases and no increases). The ratings agree with that grid: in every domain, respondents who now do the activity more in English attributed a decrease to AI more often (Spearman ρ 0.27–0.43, all $p < 0.01$). The *displacement score* is minus the mean change over the applicable domains (at least six; 104 respondents, 88 in the core sample), so higher means more AI-attributed decrease ($\alpha = 0.87$; McDonald’s $\omega = 0.87$ [0.80, 0.91] [22]). Because this consistency may partly reflect a general tendency to blame AI (common-method variance [23]), the score is read as attributed decrease, not domain loss.

Perceived difficulty without AI: “Because of AI, has each of these become harder or easier for you, when you use Arabic on your own without AI?” (recalling the everyday word, speaking one’s dialect fluently, saying what one means fully in Arabic without English words; higher = harder; $\alpha = 0.79$, $\omega = 0.81$; 63 of 105 scored 0); a fourth item, writing formal Fusha, was analysed only for the 82 who wrote Fusha before AI. *AI use*: frequency, share of work or study done with AI, breadth (number of eight uses ticked), start year, the language used to write to AI (English share, 1 = always Arabic to 5 = always English), and exposure to AI-generated content and its language. *AI-use intensity* is the mean of the standardised share and breadth ($r = 0.27$; frequency, with 79 of 105 in its top category, is descriptive only). *Perceived quality gap*: four agreement items (1 to 5), one reversed ($\alpha = 0.69$). Also asked: substitution (doing in English what one could do in Arabic, because AI helps better in English), a ninth change row on switching between Arabic and English, the same change question for social media (one item), life events, expected change in two years, and English level.

C. Analysis

Family A (H1–H2, within-person) used Wilcoxon signed-rank tests against zero, with exact sign tests for H1a and H1b, and Page’s one-sided test for the predicted order [24], with Friedman’s test and Kendall’s W . Family B regressed the displacement score on the three AI-use measures (H3, with a joint test) and perceived difficulty on the displacement score (H4), using OLS with HC3 standard errors [25], adjusted for age 25 or over and for starting English-medium study or work or moving country. Holm’s correction [26] was applied within each family (in Family B, to the three H3 coefficients and H4), and a wild bootstrap (Rademacher weights, null imposed) checked Family B [27]. H5 followed PROCESS model 4 [28] with a 100,000-resample percentile bootstrap; with cross-sectional data it does not test causal order, and at $n = 104$ it can detect medium but not small paths [29]. A Firth logistic regression [30] of any net decrease (profile penalised-likelihood tests) checked H3 in binary form. Each analytic choice is reported with its alternatives (score definition, covariates, excluded cases) as a sensitivity analysis (Table II). SPSS 23 [31] reproduced the estimates, rank tests

and reliabilities (104 statistics agree with Python [32], [33]), and Python added the HC3 errors, Holm’s correction, Page’s test and the bootstraps.

D. Machine Learning (RQ4)

The machine learning is exploratory. Targets were the number of domains with a decrease (regression), any decrease in a formal domain (work or study, or Fusha), a decrease in each domain, and three further targets (substitution, expected decrease in two years, net perceived difficulty). Features came in nested blocks: background (age 25 or over, computing field, education, English level, English-medium start or move), plus AI use (seven items), plus the quality gap. L2 logistic or ridge regression (primary), random forests and gradient boosting, with fixed settings, were evaluated by repeated 10×5 -fold cross-validation, with oversampling (SMOTENC [34]) inside the training folds only, using scikit-learn and imbalanced-learn [35], [36]. Comparisons used the Nadeau–Bengio corrected t [37], and permutation tests (500, whole pipeline refit) with Holm’s correction tested whether a model beat chance. By criteria fixed in advance, a model was usable within this sample if it was (1) above chance after Holm’s correction (within the two overall targets and within the eight domains); (2) stable over ten other random splits (mean AUC at least 0.70, none below 0.65; for a count, $Q^2 > 0$ throughout and a significant gain over the mean); (3) above chance without the age variable; and (4) at least 0.65 in balanced accuracy.

IV. RESULTS

A. Where Respondents Report Change (RQ1)

Sixty-eight of the 105 respondents (65%) reported an AI-attributed decrease in at least one domain (3.4 domains on average among them). Decreases were most common in work or study (50% [41, 59], against 3% reporting more), writing messages and posts (40%) and following Arabic content (36%), and least common with family and friends (13%) and for religious texts, where increases (12%) outnumbered decreases (9%); for Fusha, 29 of the 100 to whom it applied reported less and 11 more (Fig. 1). More than half (59 of 105, 56% [47, 65]) reported at least sometimes doing in English what they could have done in Arabic, because AI works better in English. Switching between Arabic and English while working or studying increased more often than it decreased (39 against 19, sign test $p = 0.012$).

B. Direction and Order (H1–H2)

The net change was negative for Fusha (H1a) and for work or study (H1b), and Page’s test of the predicted order was significant (H2a; Table I), carried by work or study: personal matters and family barely differed (0.14 vs. 0.10, Wilcoxon $p = 0.41$). All three held in the core sample (Holm $p \leq 0.003$). Among the 25 respondents who were studying or working and reported no new university, graduation or job, decreases in work or study still outnumbered increases (8 against 1, sign test $p = 0.039$, or 7 against 1, $p = 0.070$, without three who started studying or working in English; Wilcoxon exact

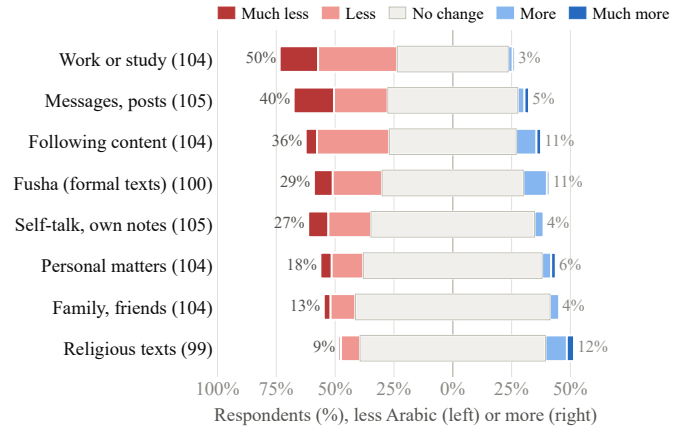


Fig. 1. AI-attributed change in Arabic use by domain (all respondents), centred on “no change”; n in parentheses excludes respondents who marked the domain not applicable.

$p = 0.12$), and work or study still showed the largest decrease ($n = 24$, Page $p = 0.011$). The decrease for Fusha was not significantly larger than for the dialect domains (H2b).

C. Predictors and Correlates (H3–H5)

The three AI-use measures jointly predicted the displacement score, but only through the English share (Table I): respondents who wrote to AI more often in English attributed more decrease to it. The English share was Holm-significant in every specification (Table II), including with English proficiency (correlated with the share, $\rho = 0.57$) or social-media attribution controlled; it held with starting university, graduating and a new job controlled ($b = 0.16$, $p < 0.001$) and in a Firth logistic regression of any net decrease ($p = 0.025$). Intensity and the quality gap were related to the score on their own ($b = 0.11$, $p = 0.035$; $b = 0.16$, $p = 0.037$; same covariates) but added nothing once the English share, with which both correlate ($\rho = 0.35$ and 0.37), was controlled. The 23 respondents aged 25 or over reported little change (15 with no decrease; $b = -0.30$, $p = 0.001$).

H4 fell short of significance ($p = 0.053$, Holm 0.16; unadjusted Spearman $\rho = 0.47$). It was positive in every specification, Holm-significant in four of the eleven in Table II, and clear in an ordinal (proportional-odds) model, which is less driven by a few large scores ($p < 0.001$). One respondent outside the core sample, whose answers ran against the trend, weakened it (Cook’s distance 0.80); no other single removal brought p below its Holm threshold of 0.017. Both measures are AI attributions from the same page, so H4 is not evidence that reduced use erodes ability. For H5, the path from displacement to difficulty was not significant ($p = 0.055$), the total effect was close to zero ($c = -0.03$), and the CI of the indirect effect reached zero, also with switching as a second mediator (exploratory). Among the 82 who wrote Fusha before AI, finding formal writing harder went with more attributed decrease ($\rho = 0.46$, $p < 0.001$; exploratory).

TABLE I
HYPOTHESIS TESTS (ALL RESPONDENTS)

Hypothesis	<i>n</i>	Result	Test	Holm <i>p</i>	Verdict
<i>Family A: direction and order of AI-attributed change</i>					
H1a Fusha	100	29 less, 11 more	$z = -3.06, r = -0.31$	0.004	supported
H1b Work or study	104	52 less, 3 more	$z = -6.00, r = -0.59$	<0.001	supported
H2a Work/study > personal > family	102	decrease 0.64, 0.14, 0.10	Page $L = 1289$, Kendall's $W = 0.28$	<0.001	supported
H2b Fusha > dialect domains	100	difference +0.08	$z = 1.21$	0.22	not supported
<i>Family B: H3 predicts the displacement score; H4 predicts difficulty from it ($n = 104$, HC3)</i>					
H3 (joint)	104	three predictors	Wald $F(3, 98) = 7.86, p < 0.001$		supported
H3a AI-use intensity		$b = -0.003 [-0.13, 0.12]$	$p = 0.96$	0.96	not supported
H3b English share of AI use		$b = 0.17 [0.09, 0.25], \beta = 0.41$	$p < 0.001$	<0.001	supported
H3c Perceived quality gap		$b = 0.06 [-0.07, 0.19]$	$p = 0.37$	0.75	not supported
H4 Difficulty without AI	104	$b = 0.37 [-0.01, 0.74]$	$p = 0.053$	0.16	not supported ^a
H5 Mediation	104	indirect 0.04 [0.00, 0.11]	percentile bootstrap		not supported ^b

Wilcoxon tests against zero, $r = z/\sqrt{n}$ with n including no-change answers; exact sign tests agree. H2a: Page's one-sided test on complete-case decrease scores (Friedman $\chi^2(2) = 57.06$), carried by work or study (personal vs. family $p = 0.41$). Family B: adjusted for age 25+ and an English-medium start or move; 95% CIs; H3 $R^2 = 0.26$; wild bootstrap Holm p : H3b 0.001, H4 0.23. ^aPositive in every specification; Holm-significant in four of the eleven in Table II. ^b100,000 resamples; lower limit -0.0001 ; the path from displacement to difficulty has $p = 0.055$.

TABLE II
ENGLISH SHARE (H3B) AND H4 UNDER ALTERNATIVE SPECIFICATIONS

Specification	English share		H4	
	<i>b</i>	Holm <i>p</i>	<i>b</i>	Holm <i>p</i>
Primary specification	0.17	<0.001	0.37	0.16
Four-domain score	0.18	<0.001	0.34	0.028
Dialect domains only	0.17	<0.001	0.41	0.004
Age as the only covariate	0.17	<0.001	0.37	0.15
No covariates	0.16	0.001	0.38	0.11
Adding English proficiency	0.15	0.042	0.32	0.38
Adding field and AI tenure	0.18	0.001	0.37	0.13
Adding social-media attribution	0.12	0.021	0.34	0.14
Without straight-line responders	0.15	<0.001	0.35	0.30
Without the most influential case	0.15	<0.001	0.52	<0.001
Core sample ($n = 88$)	0.17	<0.001	0.53	<0.001

Holm's correction across the four Family B tests in each row. Straight-line responders: "much less" on every change row (two; one has a score). Most influential case: largest Cook's distance for H4. Social-media attribution shares the AI items' format, so it also absorbs a general tendency to attribute change.

D. Social Media and Life Transitions

Respondents who reported less Arabic because of AI also reported less because of social media ($\rho = 0.55$). The single social-media rating showed more decrease than the AI ratings averaged over the domains (+0.48 vs. +0.27, scored like the displacement score; $p < 0.001$) and as much as the AI ratings for the two formal domains (+0.45, $p = 0.35$). The AI-attributed decrease was concentrated among those who reported a social-media decrease ($n = 48$ with a score; +0.54, against +0.04 for the other 56; exploratory), and the English share still predicted it with social-media attribution controlled (Table II). Life transitions coincided with the AI period: 66 of the 82 respondents under 25 started university, graduated, began a job or started studying or working in English. The 36 who started university (35 with work or study applicable) reported larger work/study decreases than the rest (-1.03 vs. -0.42 , Mann-Whitney $p < 0.001$; exploratory), and the 25

with no new university, graduation or job smaller ones (-0.28 vs. -0.73 , $p = 0.022$), although not once age was adjusted for ($p = 0.080$; 11 of the 25 were 25 or over).

E. Prediction (RQ4)

Three linear models met all four criteria (Table III): background plus AI use predicted a decrease in any formal domain (AUC 0.75, 95% CI 0.66–0.84; 0.76 over ten other splits; balanced accuracy 0.71), in work or study (0.74, 0.64–0.83; 0.74; 0.70) and in self-talk and own notes (0.73, 0.62–0.85; 0.75; 0.65), and each stayed above chance without the age variable (permutation $p = 0.018, 0.006, 0.002$). The scores rank respondents rather than estimate risk: out of fold, 75% and 72% of those scoring 0.5 or more on the two formal-domain models had reported the decrease, and 56% of those scoring 0.6 or more for self-talk (base rate 27%). Writing missed the stability criterion (0.69 across splits); personal matters, family and Fusha reached 0.68–0.69 but were not above chance after correction (Holm p 0.06–0.09). The count of domains with a decrease beat chance ($Q^2 = 0.06$, permutation $p = 0.004$) but not, significantly, the mean ($p = 0.46$).

Adding AI-use information to background never improved prediction significantly (RQ4): +0.02 [−0.07, 0.10] AUC for any formal-domain decrease, +0.01 [−0.16, 0.18] Q^2 for the count (Holm $p = 1.00$ for both), and −0.10 to +0.16 for single domains (smallest Holm p 0.27); the test could detect only gains of about 0.12 AUC. Background alone and AI use alone each reached AUC 0.70–0.73 for the formal and work/study targets, so the blocks overlap (English proficiency, a background feature, correlates with the English share). For self-talk, background alone was not above chance after correction (Holm $p = 0.072$). Random forests and gradient boosting never met the preset rule for beating the linear model. Six oversampling strategies inside the training folds gave no reliable gain (best +0.02 AUC; Holm $p = 1.00$ across 24 comparisons); four of them, applied to all respondents before cross-validation, raised the apparent AUC up to 0.89, which

measures recognition of synthetic copies, not prediction, an error behind near-perfect results elsewhere [38], [39].

V. DISCUSSION

The reported decrease centred on work or study, writing and following content, with little in family use and none, on balance, in religion. Its formal part resembles reports from Gulf universities [5], [6], but here it is attributed to AI and extends to messages and media. Work or study came first, as H2a predicts, but this may also mirror where AI is used, and the Fusha–dialect divide did not order the attributions: Fusha did not decline more than the dialect domains, religious texts, also in Fusha, showed no net decrease, and writing messages, usually in the dialect, showed the second largest. As switching between the two languages increased more often than it decreased, for some respondents AI seems to add English alongside Arabic rather than replace it.

AI is not the first technology to which these speakers attribute a move away from Arabic. Social media had moved young Arabs’ informal writing into Latin script, often mixed with English [17], [18], and respondents who reported less Arabic because of AI also reported less because of social media. The two can be read as successive stages of one technology-driven shift: social media moved informal communication away from Arabic, and AI carries the shift into work and study as well as everyday writing. The AI attribution also has a pattern of its own: in every specification, including with social-media attribution controlled, it went with the language in which people use AI more than with how much they use it. As it also went with decreases with family and in self-talk, the English share may mark a wider orientation toward English, of which AI use is one expression. The items named AI as the cause without asking respondents to set aside other causes, and the AI period also covers the move to university or work, which shapes perceived language change [20], so AI’s own share is a question for longitudinal designs. For RQ3, the decrease and perceived difficulty without AI were positively but not robustly related (H4), and mediation was not supported (H5).

The usable models identify, within this sample, who reports decreases in formal domains, work or study and self-talk, at an AUC of about 0.75 (lower 95% bounds 0.62–0.66); chosen among 13 targets on the same data, they are likely somewhat optimistic, and external validation comes next. AI use added no significant gain over background, so RQ4 is not supported, although gains of up to about 0.1 AUC cannot be excluded.

Limitations and future work. The sample, recruited through university, personal and social-media networks, is young, highly educated and computing-heavy, and people interested in the topic may have been more likely to take part, so the percentages describe this group rather than all Lebanese bilinguals. Apart from the expert review, the instrument was not back-translated or formally pretested, and change was rated retrospectively, over two to four years for most respondents (94 of 105 began regular AI use in 2024 or earlier). All measures are self-reports from one sitting, so common-method

variance may inflate their associations, H4 most. The items record Arabic as a whole: they do not separate doing an activity in English from letting AI produce the Arabic, say whether Arabic in Latin letters (Arabizi) counts, or cover French, the school language of a third of respondents. Future work can build on these results with instruments that separate these cases, larger and more varied samples, repeated measures of AI and social-media attributions with a separate rating for other causes, external validation, and records of the language people use with AI.

VI. CONCLUSION

Two thirds of the Lebanese bilinguals in this study report that generative AI has reduced their use of Arabic, most often in work or study and in writing, and those who use AI more in English report more decrease. The pattern is consistent with AI extending a shift that social media began into work, study and everyday writing. Simple models identify, with moderate accuracy and within this sample, who reports decreases in formal areas, work or study and self-talk, although AI-use information added nothing significant to background characteristics. Longitudinal designs and behavioural measures can now test how much of the reported change AI itself drives.

DATA AND CODE AVAILABILITY

The instrument, SPSS syntax and output, and group-level results are at <https://arabic-ai.hadighazi.com>, and the code is available on request. Individual responses are not shared, as participants were told results would be reported in aggregate.

APPENDIX: KEY ITEMS AS FIELDIED

<i>Because of AI, do you now do each of these less or more?</i>	بسبب الذكاء الاصطناعي، هل أصبحت تقوم بكلّ مما يلي أقل أم أكثر؟
Using Arabic in my work or studies	استخدام العربية في عملي أو دراسي
Writing messages or posts in Arabic	كتابة الرسائل أو المنشورات بالعربية
Using Arabic when I talk to myself or write my own notes	استخدام العربية عندما أحَدّ نفسي أو أكتب ملاحظاتي الخاصة
Using Arabic when talking about my personal matters	استخدام العربية في الحديث عن أموري الشخصية
Using Arabic with my family or friends	استخدام العربية مع عائلتي أو أصدقائي
Reading or writing Fusha (news, documents, articles)	قراءة الفصحى أو كتابتها (أخبار، وثائق، مقالات)
Reading or listening to religious texts	قراءة نصوص دينية أو الاستماع إليها
Following content in Arabic (videos, articles)	متابعة محتوى بالعربية (فيديوهات، مقالات)
Switching between Arabic and English while working or studying	التنقل بين العربية والإنجليزية أثناء العمل أو الدراسة

REFERENCES

- [1] K. Ahuja *et al.*, “MEGA: Multilingual evaluation of generative AI,” in *Proc. 2023 Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2023, pp. 4232–4267.
- [2] K. Kubrak, A. El-Moselhy, A. Alsulami, R. Altuwaim, H. I. Fawaz, and F. Alsaby, “Arabic prompts with English tools: A benchmark,” in *Proc. 2025 IEEE Int. Conf. Big Data (BigData)*, 2025, pp. 5452–5461.
- [3] M. Sallam *et al.*, “Language discrepancies in the performance of generative artificial intelligence models: An examination of infectious disease queries in English and Arabic,” *BMC Infectious Diseases*, vol. 24, no. 1, 2024, Art. no. 799.
- [4] A. Al-Bataineh, “Language policy in higher education in the United Arab Emirates: Proficiency, choices and the future of Arabic,” *Language Policy*, vol. 20, no. 2, pp. 215–236, 2021.
- [5] T. I. Masri, “A critical examination of the Arab undergraduate students’ perceptions of their academic Arabic proficiency in three EMI universities in the UAE,” *English Language Teaching*, vol. 12, no. 4, p. 21, 2019.

TABLE III
CROSS-VALIDATED PREDICTION (LINEAR MODELS, ALL RESPONDENTS)

Target (n yes/no; metric)	Background	+ AI use	AI use only	Δ AI use [95% CI]	Usable
Domains with a decrease (count; Q^2)	0.05	0.06	0.02	+0.01 [−0.16, 0.18]	no
Any formal-domain decrease (55/49; AUC)	0.73	0.75 [0.66, 0.84]	0.71	+0.02 [−0.07, 0.10]	yes
Work or study (52/52; AUC)	0.73	0.74 [0.64, 0.83]	0.70	+0.00 [−0.08, 0.08]	yes
Writing messages and posts (42/63; AUC)	0.71	0.70	0.62	−0.01 [−0.09, 0.06]	no
Self-talk and own notes (28/77; AUC)	0.68	0.73 [0.62, 0.85]	0.75	+0.06 [−0.04, 0.15]	yes
Fusha (29/71; AUC)	0.64	0.68	0.67	+0.03 [−0.06, 0.13]	no
Substitution (59/46; AUC)	0.56	0.61	0.62	+0.05 [−0.07, 0.18]	no
Expected decrease in two years (count; Q^2)	0.02	−0.01	−0.01	−0.03 [−0.19, 0.14]	no
Net difficulty without AI (34/71; AUC)	0.66	0.60	0.53	−0.06 [−0.16, 0.04]	no

Means over repeated 10×5-fold cross-validation; chance: AUC 0.50; $Q^2 = 1 - \text{MSE}/\text{Var}(y)$, 0 for predicting the mean. Brackets: Hanley–McNeil 95% CI [40]. Δ : with minus without AI use, Nadeau–Bengio CI. Usable: all four criteria (Section III-D). Last three rows: further targets. Other domains: AUC 0.50–0.69. Holm over 13 targets: usable 0.026.

- [6] E. Mustafawi, K. Shaaban, T. Khwailah, and K. Ata, “Perceptions and attitudes of Qatar University students regarding the utility of Arabic and English in communication and education in Qatar,” *Language Policy*, vol. 21, no. 1, pp. 75–119, 2022.
- [7] F. Esseili, “A sociolinguistic profile of English in Lebanon,” *World Englishes*, vol. 36, no. 4, pp. 684–704, 2017.
- [8] M. Mehio, “Upholding and resisting English language ideologies: Identity negotiation of five Sunni Muslims in Lebanon,” *Journal of Language, Identity & Education*, vol. 25, no. 5, pp. 2132–2146, 2026.
- [9] M. Saarela, S. Gunasekara, and P. Kumarage, “A meta-analysis of generative AI effects on language proficiency and affective–cognitive outcomes in language learning,” *Discover Computing*, vol. 29, no. 1, 2026, Art. no. 116.
- [10] A. Bouzayenne, “Language choice and AI-mediated code-switching in Tunisians’ interactions with ChatGPT: Age-related and task-domain variation,” *Dialogica*, vol. 2, no. S1, pp. 33–54, 2026.
- [11] J. A. Fishman, “Who speaks what language to whom and when?” *La Linguistique*, vol. 1, no. 2, pp. 67–88, 1965.
- [12] J. A. Fishman, “Domains and the relationship between micro- and macrosociolinguistics,” in *Directions in Sociolinguistics: The Ethnography of Communication*, J. J. Gumperz and D. Hymes, Eds. New York, NY, USA: Holt, Rinehart and Winston, 1972, pp. 435–453.
- [13] C. A. Ferguson, “Diglossia,” *Word*, vol. 15, no. 2, pp. 325–340, 1959.
- [14] A. Chaouch-Orozco and F. Martín-Villena, “Network science reveals the early signs of L1 lexical attrition: Introducing the Lexical Attrition Foundation (LeAF) framework,” *Bilingualism: Language and Cognition*, vol. 28, no. 1, pp. 43–53, 2025.
- [15] F. Gallo *et al.*, “Lexico-semantic attrition of native language: Evidence from Russian–Hebrew bilinguals,” *Language Learning*, vol. 75, no. 3, pp. 771–800, 2025.
- [16] A. Ayçiçeği-Dinn, S. Şişman-Bal, and C. L. Caldwell-Harris, “Does attending an English-language university diminish abilities in the native language? Data from Turkey,” *Applied Linguistics*, 2015, advance online publication, doi: 10.1093/applin/amv052.
- [17] H. Alghamdi and E. Petraki, “Arabizi in Saudi Arabia: A deviant form of language or simply a form of expression?” *Social Sciences*, vol. 7, no. 9, 2018, Art. no. 155.
- [18] R. M. Bianchi, “3arabizi – when local Arabic meets global English,” *Acta Linguistica Asiatica*, vol. 2, no. 1, pp. 89–100, 2012.
- [19] E. Hargittai, “Second-level digital divide: Differences in people’s online skills,” *First Monday*, vol. 7, no. 4, 2002.
- [20] M. A. Wirtz and S. Pickl, “Major life events as drivers of perceived linguistic change across adulthood,” *Language Variation and Change*, vol. 37, no. 1, pp. 87–110, 2025.
- [21] T. Schwaba, J. J. A. Denissen, M. Luhmann, C. J. Hopwood, and W. Bleidorn, “Subjective experiences of life events match individual differences in personality development,” *Journal of Personality and Social Psychology*, vol. 125, no. 5, pp. 1136–1156, 2023.
- [22] T. J. Dunn, T. Baguley, and V. Brunsden, “From alpha to omega: A practical solution to the pervasive problem of internal consistency estimation,” *British Journal of Psychology*, vol. 105, no. 3, pp. 399–412, 2014.
- [23] P. M. Podsakoff, S. B. MacKenzie, J.-Y. Lee, and N. P. Podsakoff, “Common method biases in behavioral research: A critical review of the literature and recommended remedies,” *Journal of Applied Psychology*, vol. 88, no. 5, pp. 879–903, 2003.
- [24] E. B. Page, “Ordered hypotheses for multiple treatments: A significance test for linear ranks,” *Journal of the American Statistical Association*, vol. 58, no. 301, pp. 216–230, 1963.
- [25] J. S. Long and L. H. Ervin, “Using heteroscedasticity consistent standard errors in the linear regression model,” *The American Statistician*, vol. 54, no. 3, pp. 217–224, 2000.
- [26] S. Holm, “A simple sequentially rejective multiple test procedure,” *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.
- [27] R. Davidson and E. Flachaire, “The wild bootstrap, tamed at last,” *Journal of Econometrics*, vol. 146, no. 1, pp. 162–169, 2008.
- [28] A. F. Hayes, *Introduction to Mediation, Moderation, and Conditional Process Analysis: A Regression-Based Approach*, 3rd ed. New York, NY, USA: Guilford Press, 2022.
- [29] M. S. Fritz and D. P. MacKinnon, “Required sample size to detect the mediated effect,” *Psychological Science*, vol. 18, no. 3, pp. 233–239, 2007.
- [30] D. Firth, “Bias reduction of maximum likelihood estimates,” *Biometrika*, vol. 80, no. 1, pp. 27–38, 1993.
- [31] *IBM SPSS Statistics for Windows, Version 23.0*, IBM Corp., Armonk, NY, USA, 2015.
- [32] S. Seabold and J. Perktold, “Statsmodels: Econometric and statistical modeling with Python,” in *Proc. 9th Python in Science Conf.*, 2010, pp. 92–96.
- [33] P. Virtanen *et al.*, “SciPy 1.0: Fundamental algorithms for scientific computing in Python,” *Nature Methods*, vol. 17, no. 3, pp. 261–272, 2020.
- [34] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, “SMOTE: Synthetic minority over-sampling technique,” *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.
- [35] F. Pedregosa *et al.*, “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [36] G. Lemaitre, F. Nogueira, and C. K. Aridas, “Imbalanced-learn: A Python toolbox to tackle the curse of imbalanced datasets in machine learning,” *Journal of Machine Learning Research*, vol. 18, no. 17, pp. 1–5, 2017.
- [37] C. Nadeau and Y. Bengio, “Inference for the generalization error,” *Machine Learning*, vol. 52, no. 3, pp. 239–281, 2003.
- [38] G. Vandewiele *et al.*, “Overly optimistic prediction results on imbalanced data: A case study of flaws and benefits when applying over-sampling,” *Artificial Intelligence in Medicine*, vol. 111, 2021, Art. no. 101987.
- [39] M. S. Santos, J. P. Soares, P. H. Abreu, H. Araujo, and J. Santos, “Cross-validation for imbalanced datasets: Avoiding overoptimistic and overfitting approaches,” *IEEE Computational Intelligence Magazine*, vol. 13, no. 4, pp. 59–76, 2018.
- [40] J. A. Hanley and B. J. McNeil, “The meaning and use of the area under a receiver operating characteristic (ROC) curve,” *Radiology*, vol. 143, no. 1, pp. 29–36, 1982.